1 THE T-BLOCK THEOREM

1.1 At the tower boundary, sheet identities split by large-exponent support, forcing Koszul antisymmetry

Status: pencil-proven for all v 2; pending the project's P0 program. Self-contained; the v=1 boundary is explicitly excluded (and handled by the sealed A(3) egg).

1.1.1 0. Why this theorem matters (Quanta paragraph)

The Recognition Theorem tamed the *middle* of a Frobenius-tower problem by observing that below the threshold `q`, the large power `s_i^q` cannot interfere with itself: monomial supports stay disjoint and the tower drops out. But at the boundary — coefficient degrees `q  e < q+4`, where the ideal `m^{[q]}` first bites — that disjointness begins to fail, and this is exactly where q-freeness genuinely breaks. The T-Block Theorem is the precise anatomy of that boundary: it shows the sheet identities still split into disjoint monomial *classes* (indexed by which coordinates carry a large exponent), and that the only surviving cross-terms are forced to be **antisymmetric** — the fingerprint of a Koszul syzygy `s_i^q s_j^q - s_j^q s_i^q`. The wall, up close, is made of Koszul relations with tiny cofactors.

1.1.2 1. Setup

`S = F [s ..s ]`, `E = (e ,e ,e ) =  _J I_J` (the 15 matching-sheets of K). For a sheet `J` with pair coordinates `u ,u ,u` (via `s_b = −s_a` on each pair), a form restricts to `F [u ,u ,u ]`. Fix `q = 3^v` and `f  {0,1,2,3}`; consider `  = (  ..  )` homogeneous of degree `q+f` with `Σ  s ^q  E`, i.e. vanishing on every sheet.

On a sheet, set `Δ_k = ( _{a_k} −  _{b_k})|_{L_J}`, so the sheet identity is `Σ_{k=1}^3 Δ_k u_k^q = 0`. Since `deg Δ_k = q+f < 2q`, each monomial of `Δ_k` has **at most one** exponent `q`; split canonically `Δ_k = Δ_k^  + Σ_l u_l^q  _{k,l}`, with `Δ_k^` the reduced part (all exponents `< q`) and `deg  _{k,l} = f  3`.

1.1.3 2. The theorem

> **Theorem (T-Block).** Let `q  9` and `f  3`. Then the sheet identity `Σ_k Δ_k u_k^q = 0` forces, on every sheet: **(i)** `Δ_k^  = 0` for each `k` (the reduced part of every pair-difference vanishes); **(ii)** ` _{k,k} = 0` (no deep-diagonal `u_k^{2q}` term); **(iii)** ` _{l,k} = − _{k,l}` for `k  l` (**Koszul antisymmetry**), with cofactor degree `f  3`.

1.1.4 3. Proof

Substitute the split into `Σ_k u_k^q Δ_k = 0`: `Σ_k u_k^q Δ_k^  + Σ_{k,l} u_k^q u_l^q  _{k,l} = 0`. Index each monomial of the left side by its **set `T` of coordinates carrying an exponent `q`**. Because the total degree is `2q+f < 3q` (using `q  9 > f`), no monomial can have three large exponents, so `|T|   2`. The three resulting classes are disjoint by exponent range: - **`|T| = {k}` (one large exponent, in `[q, 2q-1]`):** contributed only by `u_k^q Δ_k^ ` (exponent of `u_k` is `q + (\text{that of }Δ_k^ ) < 2q`). Vanishing of this class `  Δ_k^  = 0` — (i). - **`|T| = {k}` with the `u_k`-exponent `  2q`:** contributed only by `u_k^q · u_k^q  _{k,k} = u_k^{2q} _{k,k}`. Vanishing `  _{k,k} = 0` — (ii). - **`|T| = {k,l}, k   l`:** contributed only by `u_k^q u_l^q( _{k,l} +  _{l,k})`. Vanishing `  _{k,l} +  _{l,k} = 0` — (iii). No class overlaps another (their large-exponent patterns differ), so each vanishes independently.

1.1.5 4. The boundary case v = 1 (why q   9 is sharp)

At `q = 3`, the top collar degree gives total degree `2q+f = 6+f`, which can reach `3q = 9` when `f = 3`; then `|T| = 3` becomes possible and the class separation fails. This is not a defect but a genuine feature of the smallest level: `v = 1` is handled separately by the sealed direct computation `A(3) = 141 = P(3)`, and every statement here is quantified `v   2` (`q   9`), exactly matching the regime the tower argument needs.

1.1.6 5. What it buys (Quanta close)

The theorem reduces every boundary syzygy to: reduced parts obeying the *main-body* conditions (so they lie in the Recognition module `M`, of known q-free dimension), plus a package of **antisymmetric** cross-data   of degree `  3`. Antisymmetry is the signature of the Koszul syzygies `s_i^q s_j^q = s_j^q s_i^q`; the remaining work (bounding how much such data can appear) becomes a *fixed*, finite linear-algebra problem — the collar rank certificate. The tower wall, resolved at the boundary, is Koszul all the way down.
